

Employment

Product Engineer Intern Isomer AI May 2025 – Present

- Ship 10+ production AI features end-to-end: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST API integrations; 394 commits across 7 production deployments.
- Own backend stack (PostgreSQL, Redis, Auth0 RBAC, Docker, Terraform, Sentry); debug and resolve production incidents.

Graduate Research Assistant NC State University April 2025 – Present

- Deploy cloud genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets.
- Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

Software Developer Intern Mphasis (HP Company) Jun 2024 – Aug 2024

- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%.
- Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.

AI/ML Intern Capgemini Jun 2023 – Aug 2023

- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%.
- Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.

Education

Raleigh, NC North Carolina State University Aug 2024 – May 2026

- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.

Mumbai, India Mumbai University Aug 2020 – May 2024

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.

Projects

KubeServe Kubernetes LLM Inference Platform 2026

- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Instrumented OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.

AlgoPulse Agentic AI Platform 2025

- Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load.
- Replaced API polling with Supabase Edge Functions for real-time AI streaming; eliminated state loss under peak traffic.

LLM Eval Harness Independent Project 2025

- Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, reasoning, and code generation tasks.
- Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap CIs and McNemar tests across 6 metrics.

Demand Paging Xinu OS – Systems Programming 2025

- Implemented 8-store virtual memory system for Xinu OS in C: page tables, demand paging, Second-Chance replacement, TLB fault handler; cut page fault rate by 25%.
- Engineered 7+ kernel syscalls (address translation, TLB, frame allocation); improved memory throughput by 15%.

Languages and Technologies

- **Languages:** Python, C, TypeScript, JavaScript, Java, SQL, Bash
- **Systems & AI Infra:** Linux, Kubernetes, KEDA, Docker, OpenTelemetry, Prometheus, Grafana, k6
- **Backend & Data:** FastAPI, Node.js, PostgreSQL, Redis, Kafka, Temporal.io, REST APIs, Terraform
- **AI/ML:** Claude/GPT-4o/Gemini APIs, LangChain, RAG, LLM evaluation, prompt engineering

Additional Experience and Awards

- **IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- **National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- **Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- **Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog